



9026 - An extraordinary $z=4.4$ starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI MRS				
	1	17940-MRS-long+medium	MIRI Medium Resolution Spectroscopy	(1) GOODS-N-17940
	2	17940-MRS-long+medium-background	MIRI Medium Resolution Spectroscopy	(2) GOODS-N-17940-background

ABSTRACT

NIRSpec observations of $z=6-12$ galaxies have revealed many unexpected properties of Cosmic Dawn sources that are unseen in the local Universe, including anomalous chemical abundance patterns (high N/O and He/H at low O/H) and high gas densities. The physical interpretation of these properties has not reached a consensus, and debate is ongoing about their ionizing sources, formation histories, and enrichment pathways. A

difficulty in unambiguously interpreting such spectra is that they typically do not detect extremely faint features or long-wavelength lines that can break degeneracies. We propose to obtain MIRI medium resolution spectroscopy of a starburst galaxy at $z=4.4$, GOODS-N-17940, that shares many features with $z>6$ sources, including high emission line equivalent widths, intense star formation, a lack of a 2175 Angstrom bump in the dust curve, and notably high He/H and super-solar N/O at sub-solar O/H and C/O. This target has an exquisitely detailed rest-UV+optical NIRS spec spectrum with over 70 detected emission and absorption lines from the Cycle 1 AURORA program, with which we can study the physical drivers of these properties in much greater detail than is possible at $z>6$. Adding the rest-frame near-infrared HeI, Paschen-series HI, [FeII], and PAH 3.3-micron lines covered by MIRI MRS will solve many mysteries in the physical interpretation of 17940's line emission, conclusively identifying the cause of its chemical abundance pattern, identifying the nature of its ionizing source, providing new constraints on its gas density, and revealing its dust composition. These results will enhance our ability to robustly interpret the spectra of Cosmic Dawn galaxies.

OBSERVING DESCRIPTION

The goal of this program is to observe a $z=4.4$ starburst galaxy, GOODS-N-17940, with MIRI medium resolution spectroscopy to detect the HeI1.083um, [FeII]1.257um, Paschen-beta, Paschen-alpha, and the 3.3-micron PAH feature, enabling improved constraints on chemical abundance patterns, gas density, ionization mechanism, and dust composition. The target has similar properties to $z>6$ galaxies and thus will improve our understanding of why $z>6$ sources show unusual properties including elevated N/O and He/H and high gas densities.

At the redshift of 17940, $z=4.411$, [FeII]1.257um, Paschen-beta, Paschen-alpha, and the 3.3-micron PAH line all fortuitously fall within the long(C) sub-band in Channels 1-3 of the MIRI IFU. The 3.3-micron PAH line is expected to be the weakest feature and thus constrains our required exposure time. We predicted its flux using a $z=0$ relation between SFR and 3.3-micron PAH luminosity, and set our required depth to detect this feature at ~ 5 sigma if it is 5 times fainter than the $z=0$ relation predicts. In 17940, it is expected that the PAH feature may be weaker than locally due to a lack of a 2175 Angstrom bump in the dust curve, low carbon abundance, and strong UV radiation fields. Inputting this 3.3-micron PAH flux limit into the MIRI MRS ETC, we found that we can detect it with sufficient S/N in an exposure time of 23890 seconds (~ 6.6 hours) in the long(C) sub-band. This is achieved using 20 groups/integration, 6 integrations/exposure, 1 exposure per dither, and a 4-point dither pattern to sample the PSF. This 4-point dither is repeated twice to achieve the required exposure time.

The HeI1.083um line flux falls in the medium(B) sub-band. Its flux was predicted from measured rest-optical helium line fluxes assuming the low-density case where HeI1.083um is the faintest. To detect HeI1.083um at ≥ 5 sigma, the ETC indicates we need 1911 seconds (~ 0.5 hours) of total exposure time. This is achieved 20 groups/integration, 1 integration/exposure, 1 exposure per dither position, and a 4-point dither pattern to sample the PSF. The medium(C) dither is only executed one time to achieve the required total exposure time.

We use simultaneous MIRI imaging to improve the astrometric reconstruction of the data cubes. Since we are executing 3 dither sequences (2 at ~3.3 hours each in long(C) and 1 at ~0.5 hours in medium (B)), we have the opportunity to use 3 different filters for simultaneous MIRI imaging. We choose to use the F770W and F1130W filters for the 2 long(C) dither sequences, and F1000W for the medium(B) dither sequence. These shorter-wavelength MIRI filters do not have any cross-talk with the IFU since the PSF is smaller than at $\lambda > 12$ microns. Since our target is in the JADES NIRCам imaging field that has extensive multi-band HST and JWST/NIRCам imaging, the addition of this deep MIRI imaging in 3 filters will enable new science cases in combination with other JWST data.

Because our target (spatial extent ~1 arcsecond) is much smaller than the smallest field of view of the MIRI IFU (Channel 1; ~3.5 arcseconds), we do not require a target acquisition and the JWST pointing accuracy is sufficient. We have therefore selected 'NONE' for target acquisition. The SLOWR1 readout pattern is used for all observations to prevent data volume excess. The 4-point dither adopted is optimized for extended sources.

Proposal 9026 - Targets - An extraordinary $z=4.4$ starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GOODSN-17940	RA: 12 36 35.4832 (189.1478467d) Dec: +62 13 50.04 (62.23057d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES				
Fixed Targets	(2)	GOODSN-17940-background	RA: 12 36 35.4832 (189.1478467d) Dec: +62 13 50.04 (62.23057d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES				

Proposal 9026 - Observation 1 - An extraordinary z=4.4 starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Observation	Proposal 9026, Observation 1: 17940-MRS-long+medium												Fri Jan 30 21:00:11 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[17940-MRS-long+medium-background (Obs 2)]												
Diagnosics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	GOODSN-17940	RA: 12 36 35.4832 (189.1478467d) Dec: +62 13 50.04 (62.23057d) Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	1	LONG(C)	MRSLONG		SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	1	LONG(C)	MRSSHORT		SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	2		IMAGER	F770W	SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	2	LONG(C)	MRSLONG		SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	2	LONG(C)	MRSSHORT		SLOWR1	20	6	1	Dither 1	4	24	11944.96	
	3		IMAGER	F1000W	SLOWR1	20	1	1	Dither 1	4	4	1911.194	
	3	MEDIUM(B)	MRSLONG		SLOWR1	20	1	1	Dither 1	4	4	1911.194	
	3	MEDIUM(B)	MRSSHORT		SLOWR1	20	1	1	Dither 1	4	4	1911.194	

Proposal 9026 - Observation 1 - An extraordinary $z=4.4$ starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 9026 - Observation 2 - An extraordinary z=4.4 starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Observation	Proposal 9026, Observation 2: 17940-MRS-long+medium-background												Fri Jan 30 21:00:11 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [17940-MRS-long+medium (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	GOODSN-17940-background	RA: 12 36 35.4832 (189.1478467d) Dec: +62 13 50.04 (62.23057d) Equinox: J2000				Epoch of Position: 2000							
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		Imager				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	2-Point				BACKGROUND				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F1130W	SLOWR1	20	1	1	Dither 1	2	2	955.597		
	1	LONG(C)	MRSLONG		SLOWR1	20	1	1	Dither 1	2	2	955.597		
	1	LONG(C)	MRSSHORT		SLOWR1	20	1	1	Dither 1	2	2	955.597		
	2		IMAGER	F770W	SLOWR1	20	1	1	Dither 1	2	2	955.597		
	2	LONG(C)	MRSLONG		SLOWR1	20	1	1	Dither 1	2	2	955.597		
	2	LONG(C)	MRSSHORT		SLOWR1	20	1	1	Dither 1	2	2	955.597		
	3		IMAGER	F1000W	SLOWR1	20	1	1	Dither 1	2	2	955.597		
	3	MEDIUM(B)	MRSLONG		SLOWR1	20	1	1	Dither 1	2	2	955.597		
	3	MEDIUM(B)	MRSSHORT		SLOWR1	20	1	1	Dither 1	2	2	955.597		

Proposal 9026 - Observation 2 - An extraordinary $z=4.4$ starburst as a detailed astrophysical laboratory to solve Cosmic Dawn mysteries

Special Requirements	Aperture PA Range 14.83544897 to 21.83544897 Degrees (V3 10.0 to 17.0)
	Aperture PA Range 28.83544897 to 32.83544897 Degrees (V3 24.0 to 28.0)
	Aperture PA Range 41.83544897 to 51.83544897 Degrees (V3 37.0 to 47.0)
	Aperture PA Range 55.83544897 to 56.83544897 Degrees (V3 51.0 to 52.0)
	Aperture PA Range 59.83544897 to 62.83544897 Degrees (V3 55.0 to 58.0)
	Aperture PA Range 70.83544897 to 75.83544897 Degrees (V3 66.0 to 71.0)
	Aperture PA Range 82.33544897 to 83.83544897 Degrees (V3 77.5 to 79.0)
	Aperture PA Range 89.33544897 to 95.33544897 Degrees (V3 84.5 to 90.5)
	Aperture PA Range 98.33544897 to 99.83544897 Degrees (V3 93.5 to 95.0)
	Aperture PA Range 105.53544897 to 122.83544897 Degrees (V3 100.7 to 118.0)
	Aperture PA Range 126.33544897 to 145.83544897 Degrees (V3 121.5 to 141.0)
	Aperture PA Range 150.83544897 to 157.83544897 Degrees (V3 146.0 to 153.0)
	Aperture PA Range 165.83544897 to 174.83544897 Degrees (V3 161.0 to 170.0)
	Aperture PA Range 180.83544897 to 184.83544897 Degrees (V3 176.0 to 180.0)
	Aperture PA Range 189.33544897 to 190.33544897 Degrees (V3 184.5 to 185.5)
	Aperture PA Range 193.83544897 to 203.33544897 Degrees (V3 189.0 to 198.5)
	Aperture PA Range 209.83544897 to 210.83544897 Degrees (V3 205.0 to 206.0)
	Aperture PA Range 219.83544897 to 220.83544897 Degrees (V3 215.0 to 216.0)
	Aperture PA Range 230.83544897 to 233.13544897 Degrees (V3 226.0 to 228.3)
	Aperture PA Range 248.83544897 to 252.33544897 Degrees (V3 244.0 to 247.5)
	Aperture PA Range 258.33544897 to 277.33544897 Degrees (V3 253.5 to 272.5)
	Aperture PA Range 281.33544897 to 297.33544897 Degrees (V3 276.5 to 292.5)
	Aperture PA Range 304.83544897 to 315.83544897 Degrees (V3 300.0 to 311.0)
	Aperture PA Range 325.33544897 to 341.33544897 Degrees (V3 320.5 to 336.5)
	Aperture PA Range 345.33544897 to 351.83544897 Degrees (V3 340.5 to 347.0)
	Aperture PA Range 358.83544897 to 7.83544897 Degrees (V3 354.0 to 3.0)
	Sequence Observations 1, 2, Non-interruptible